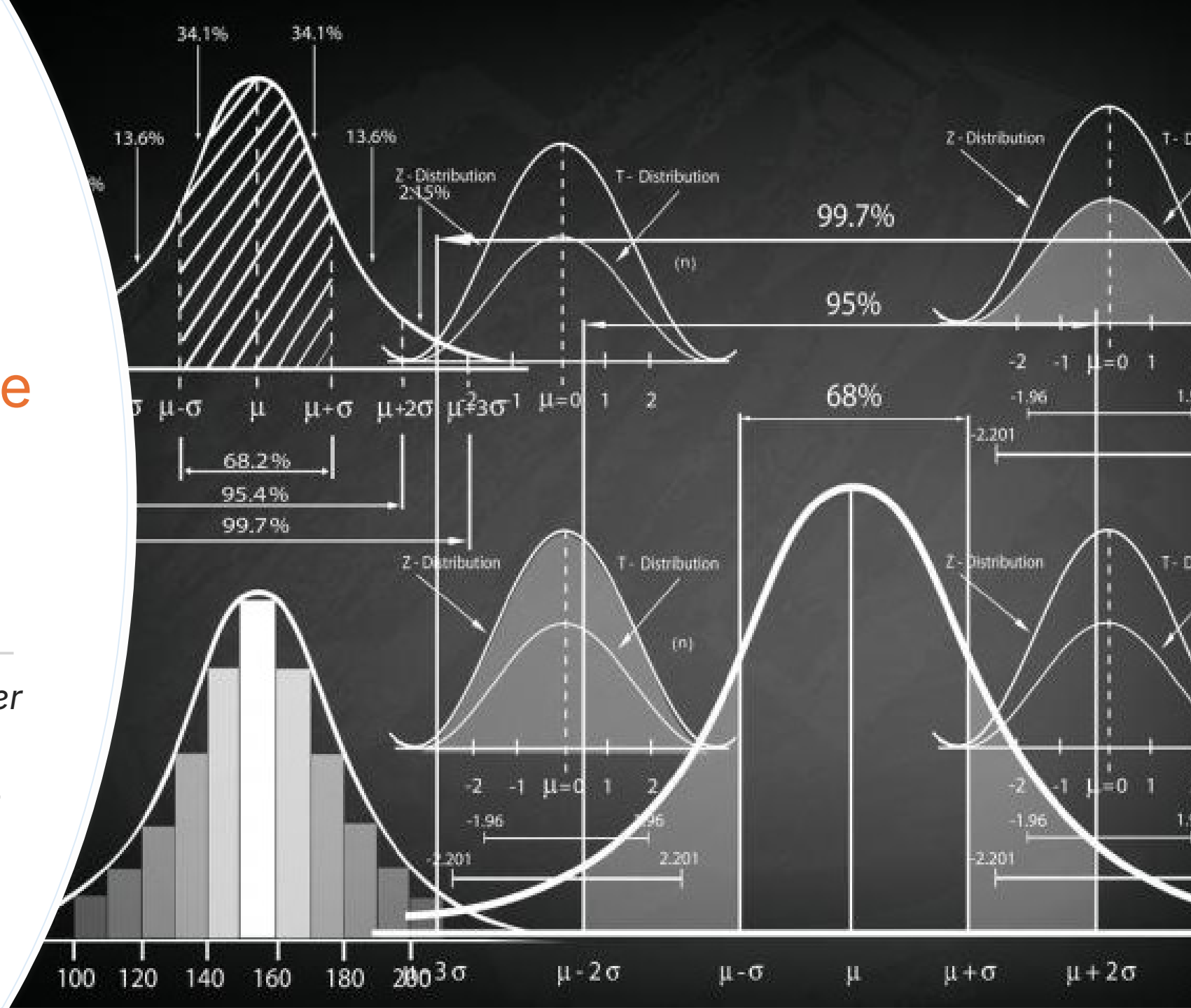
Comparing Mutual Information in Finite Samples

Can MI-specific degrees of freedom improve comparisons between populations?

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Motivation: comparing MI across populations

Mutual information measures how much observing one variable reduces uncertainty about another. Different sample estimates can arise from sampling noise.

Null hypothesis $H_0 : I(P) = I(Q)$

Online shoppers

P: new visitors
1,694 sessions

corrected MI estimate
0.0043 nats

Q: returning visitors
10,551 sessions

corrected MI estimate
0.0179 nats

Variables: month of visit and purchase outcome.

Statistical task

Draw a count table from each population

Estimate MI from each table

Assess whether sampling noise can explain the difference

Does month–purchase MI differ between the populations?

Month × purchase tables (10×2); corrected MI estimates in nats. Data: Sakar and Kastro (2018). Bossomaier et al. (2016)

Literature review: foundations and the MI-specific step

Existing results provide MI estimates, a normal-reference comparison and moment matching.
This project derives the MI variance sensitivity needed for its degrees of freedom.

	MI estimation and uncertainty	*Normal Wald comparison*	*Welch–Satterthwaite moment matching*
Idea	Estimate MI and first-order sampling variance	Divide the estimated difference by its standard error	Match variance moments to choose degrees of freedom
Key work	Moddemeijer (1989); Paninski (2003)	Mora and Ruiz-Castillo (2009)	Satterthwaite (1946); Welch (1947); Hutcheson (1970); Pan and Wall (2002)
Required step	Track variation in the variance estimate itself	Assess the fixed normal cutoff in finite samples	Calculate the MI-specific variance sensitivity

Contribution: Derive that sensitivity and evaluate the resulting degrees of freedom.

Thesis Sections 2.2–2.5 and 3.2–3.3; full references in B13–B14

Method: one statistic, two reference distributions

Both methods use the same estimated MI difference and standard error. Expanded Welch uses estimated degrees of freedom for a Student t reference.

Shared statistic

$$T = \frac{\widehat{\Delta}_{BC}}{\widehat{SE}}, \qquad \widehat{SE}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}$$

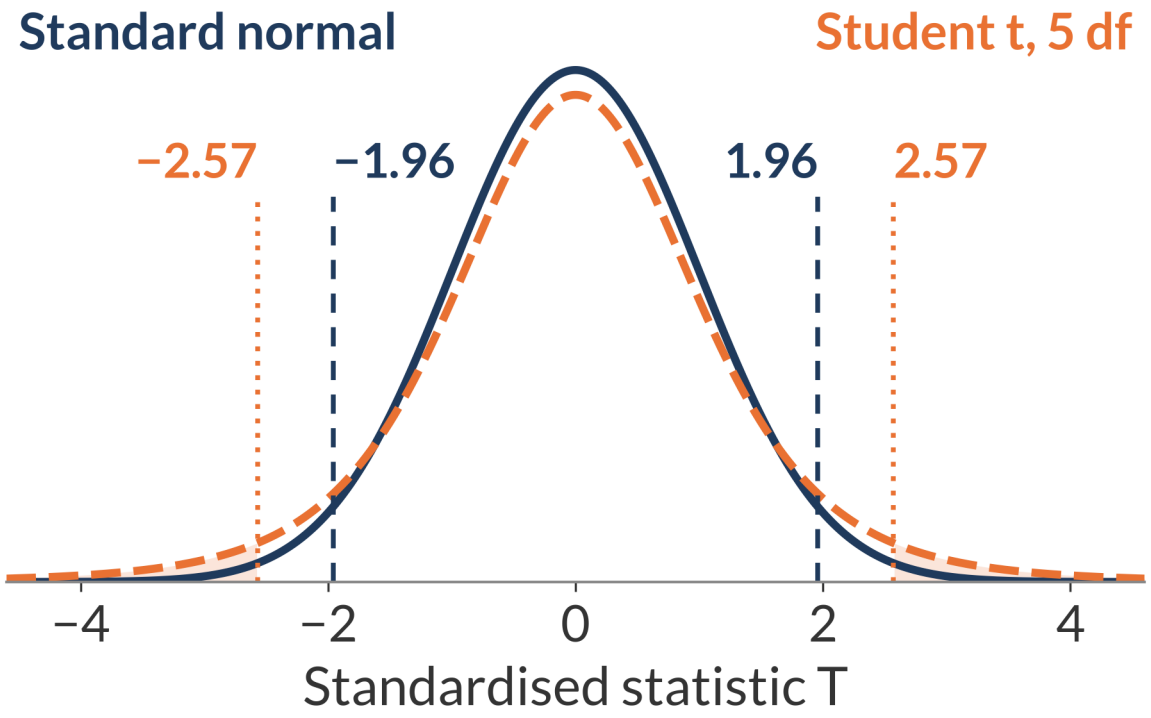
$\widehat{\Delta}_{BC}$: estimated $I(P) - I(Q)$, after leading bias correction.

Normal Wald

Standard normal distribution; two-sided 5% cutoff about **±1.96**.

Expanded Welch

Student t distribution with estimated degrees of freedom; **a larger cutoff**.



Student t distribution with 5 degrees of freedom, for illustration; cutoff about ±2.57.

Thesis Chapters 3–4, Figure 2.1. Additional validity rules: Slide 6 and B3

Approach and contribution

Tracking how the pointwise-MI variance changes with the table gives the sampling-variance input for MI-specific degrees of freedom.

Derivation chain

Cell probabilities change
Move a small amount of probability towards one cell.

→

Margins and pointwise MI change
Track their joint effect.

→

Variance sensitivity, derived here
$g_P(x, y)$ describes the change in $V(P)$.

→

Degrees of freedom
Use the spread of those sensitivities.

Result and approximations

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P} \quad \Longrightarrow \quad \hat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\hat{\tau}^2(P)}$$

$\tau^2(P)$ is the variance of $g_P(X, Y)$. Repeat for Q and combine using Welch–Satterthwaite.

Two approximations evaluated

1. First-order sampling expansion
2. Scaled chi-squared moment matching

Thesis Sections 4.1–4.3; exact derivative in B1, moment matching in B2

Evaluation design

Known populations separate approximation accuracy, false positives and power. Both tests receive the same sampled table pairs.

Research questions

1. **RQ1: Approximation accuracy**
   Sampling moments of $\hat{V}$ and component df, including near independence
2. **RQ2: False positives**
   Equal MI at 0.005, 0.02 and 0.30 nats
3. **RQ3: Power**
   P MI 0.02; MI-gap magnitudes 0.005, 0.01 and 0.02 nats

Shared simulation grid

- Four sizes: 2×2, 4×4, 8×8, 16×16
- Uniform, same-skew and different-skew margins
- Ten sample sizes: 10 to 50,000 observations per population
- 100,000 independent table pairs per setting

Model: copying construction: choose margins and solve for target MI.

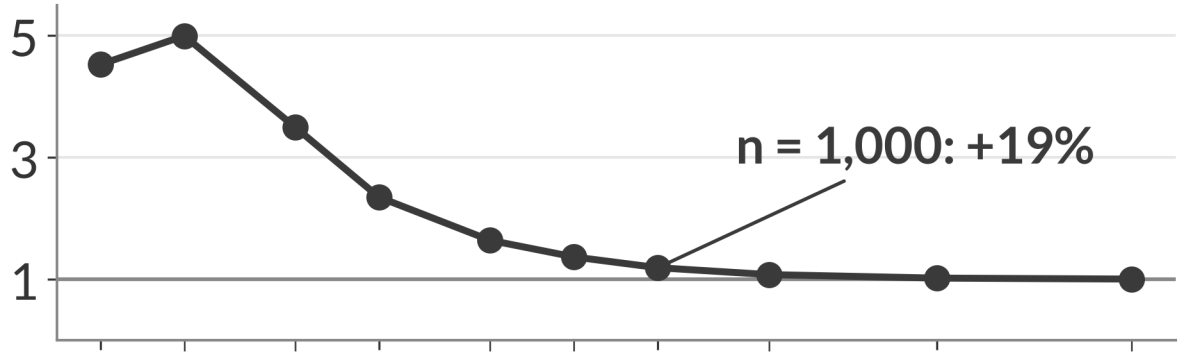
Reporting: usual 0.05 p-value thresholds. Invalid outputs count as non-rejections; validity is reported separately.

Morris et al. (2019). Thesis Chapter 5, Tables 5.1–5.3; exact vectors, sample sizes and rules in B3

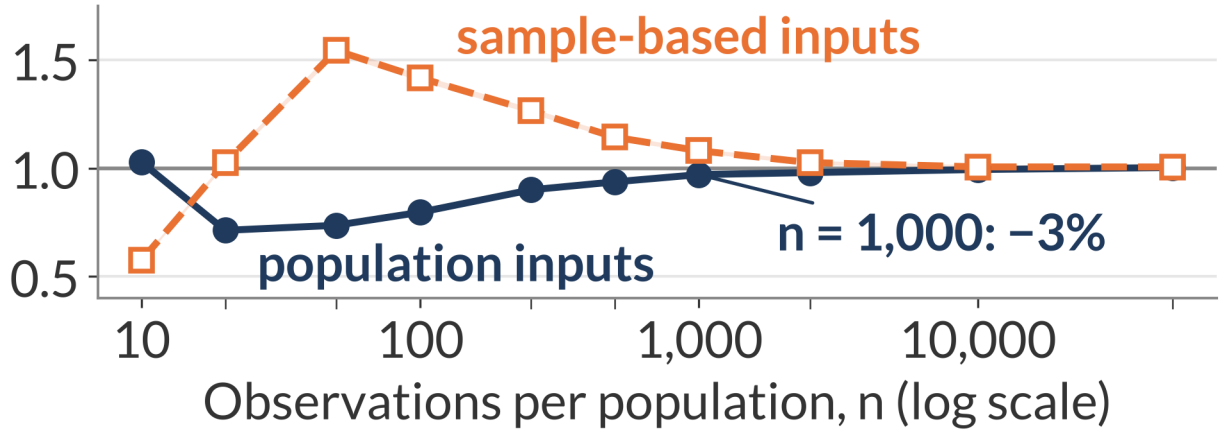
Results: approximation accuracy

The mean of $\hat{V}$ can remain inflated even when its population first-order sampling-variance prediction is accurate.

Mean of $\hat{V}$ / population V



Predicted / observed sampling variance of $\hat{V}$



Different-skew P · 4×4 · population MI 0.02 nats

Mean: rare cells can inflate it.

Newly observed rare cells can have large pointwise-MI deviations, whose squares raise $\hat{V}$.

Sampling variance: linearisation can miss large changes.

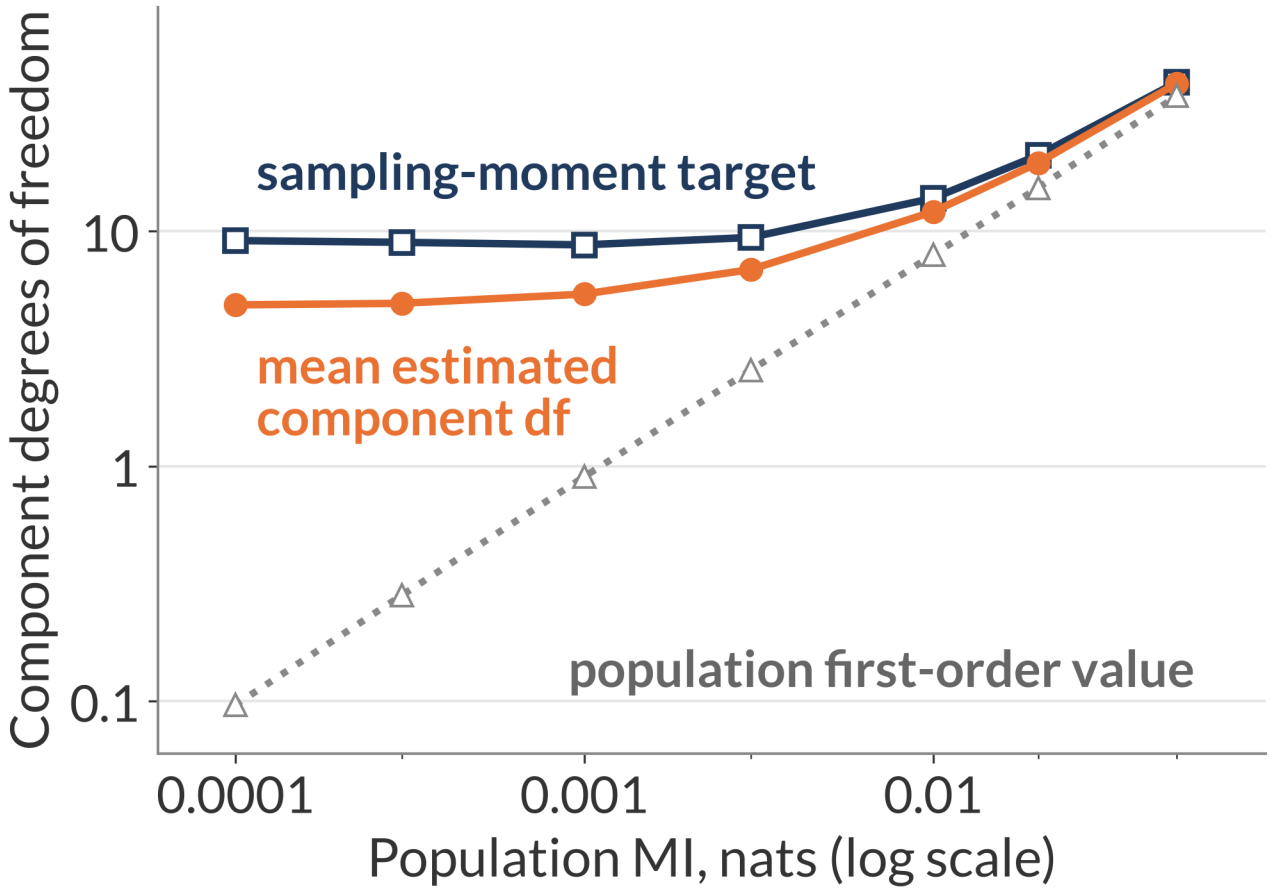
Rare cells appear in some samples and not others; sample-based sensitivities can also overpredict the resulting variation.

At n = 1,000: mean 19% high; population variance prediction 3% low. Both moments enter the degrees of freedom.

Thesis Sections 6.1.1–6.1.2, Figures 6.1–6.2. Bands: approximate 95% Monte Carlo intervals. Other sizes and Q: B4

Results: near independence

Near independence, sampling noise keeps the mean estimated component degrees of freedom positive while the population first-order value becomes very small.



Different-skew P · 4×4 · n = 1,000 fixed

Population first-order value

Insert known population inputs into the component formula. Near independence, its local value is approximately n × population MI.

Sample estimate

Quadratic sampling errors create apparent association, which remains in the sample-based calculation.

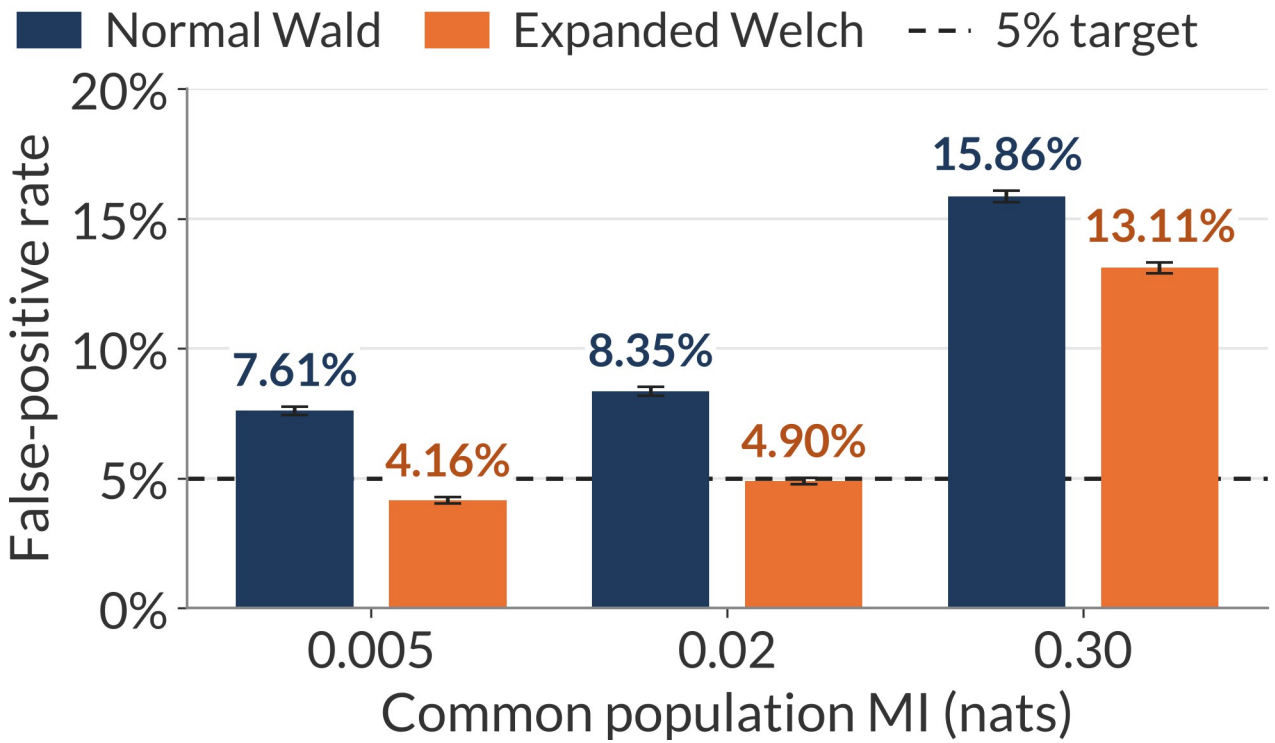
At MI 0.0001: mean estimated component df ≈4.9; sampling-moment target ≈9.1; population first-order value **≈0.1**

Thesis Section 6.1.3, Figure 6.3; Appendix A.2. Bands: approximate 95% Monte Carlo intervals. Local calculation: B5

Results: false positives across common MI values

A larger cutoff brings some excessive false-positive rates closer to 5%, but accuracy at one association level does not persist across common MI values.

Uniform 4×4 · n = 20 per population · equal MI in P and Q



Common MI 0.02: closer to 5%

Expanded Welch lowers the rate from 8.35% to 4.90%.

Common MI 0.30: still above 5%

Expanded Welch still rejects in 13.11% of repetitions.

Interpretation: The larger cutoff reduces rejection, while the shared statistic can still have the wrong sampling distribution.

Nominal target: 5%. Both tests are valid in every repetition in these examples.

Slices of the balanced null grid. Whiskers: pointwise 95% Monte Carlo intervals. Thesis Section 6.2.1, Figure 6.4; full grids in B6–B8

Results: why bias changes rejection

Similar errors in estimated uncertainty accompany opposite false-positive behaviour when bias in the MI difference changes.

16×16 · n = 1,000 per population · equal MI 0.02 nats

Margins	Variance ratio	Bias / SD	Normal Wald	Expanded Welch
Same skew	1.93	≈0	0.60%	0.53%
Different skew	1.94	**2.06**	26.01%	24.84%

Variance ratio: average squared estimated standard error / observed variance of the estimated MI difference.

Same skew

The oversized denominator suppresses the statistic; expected MI biases cancel.

Different skew

The estimated difference is shifted away from zero by unequal residual biases.

Conclusion: Changing the cutoff leaves the biased estimated difference unchanged. Both tests are valid in every repetition here.

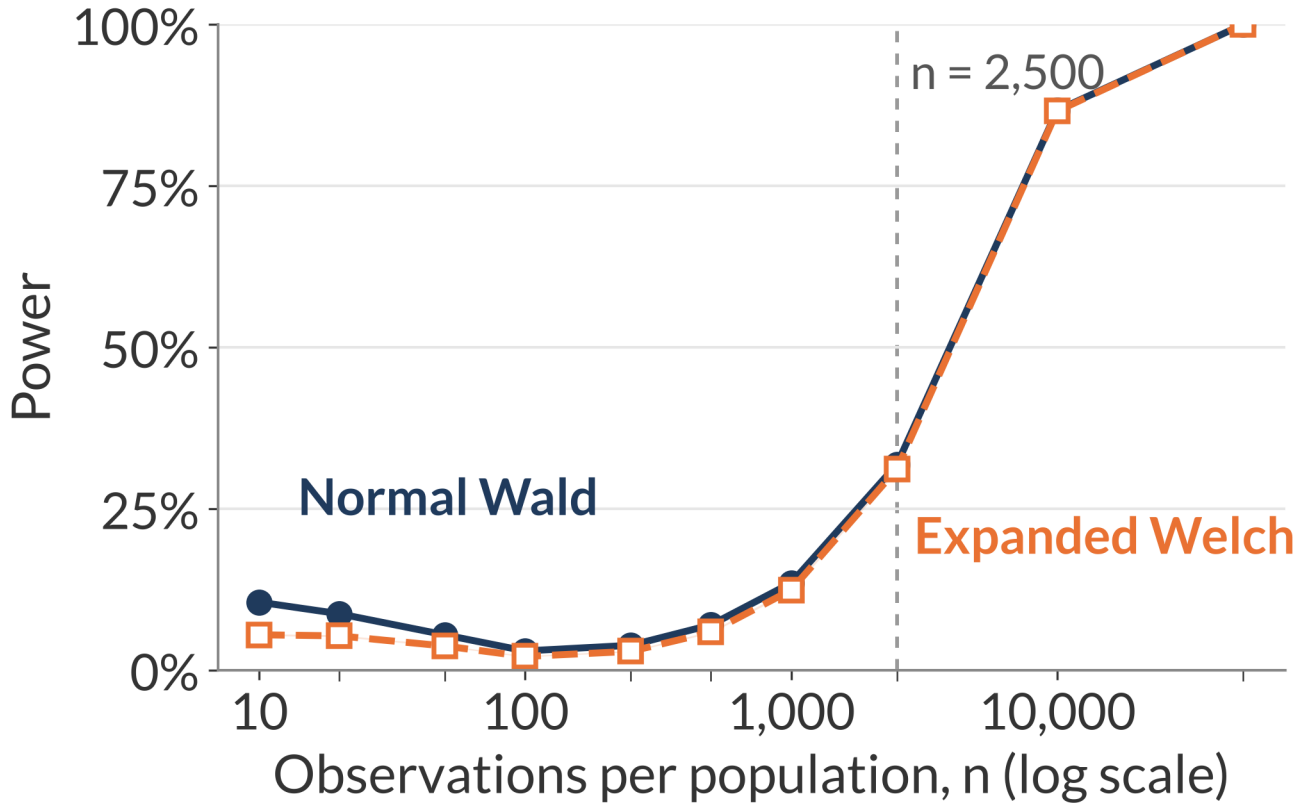
Thesis Section 6.2.4, Table 6.1; current null rates and diagnostics. Full different-skew grid: B8

Results: power across MI differences

With sufficient observations, larger MI differences become easier to detect. Expanded Welch's larger cutoff lowers or preserves rejection.

Uniform 4×4 · population MI in P = 0.02 nats

MI-gap magnitude 0.01 nats: power against sample size



Snapshot at n = 2,500

MI-gap magnitude (nats)	Normal Wald	Expanded Welch
0.005	11.60%	11.14%
0.01	31.85%	31.14%
0.02	79.24%	78.74%

Power: probability of rejecting equal MI when population MI differs. Rates use the usual 0.05 p-value threshold.

Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections.

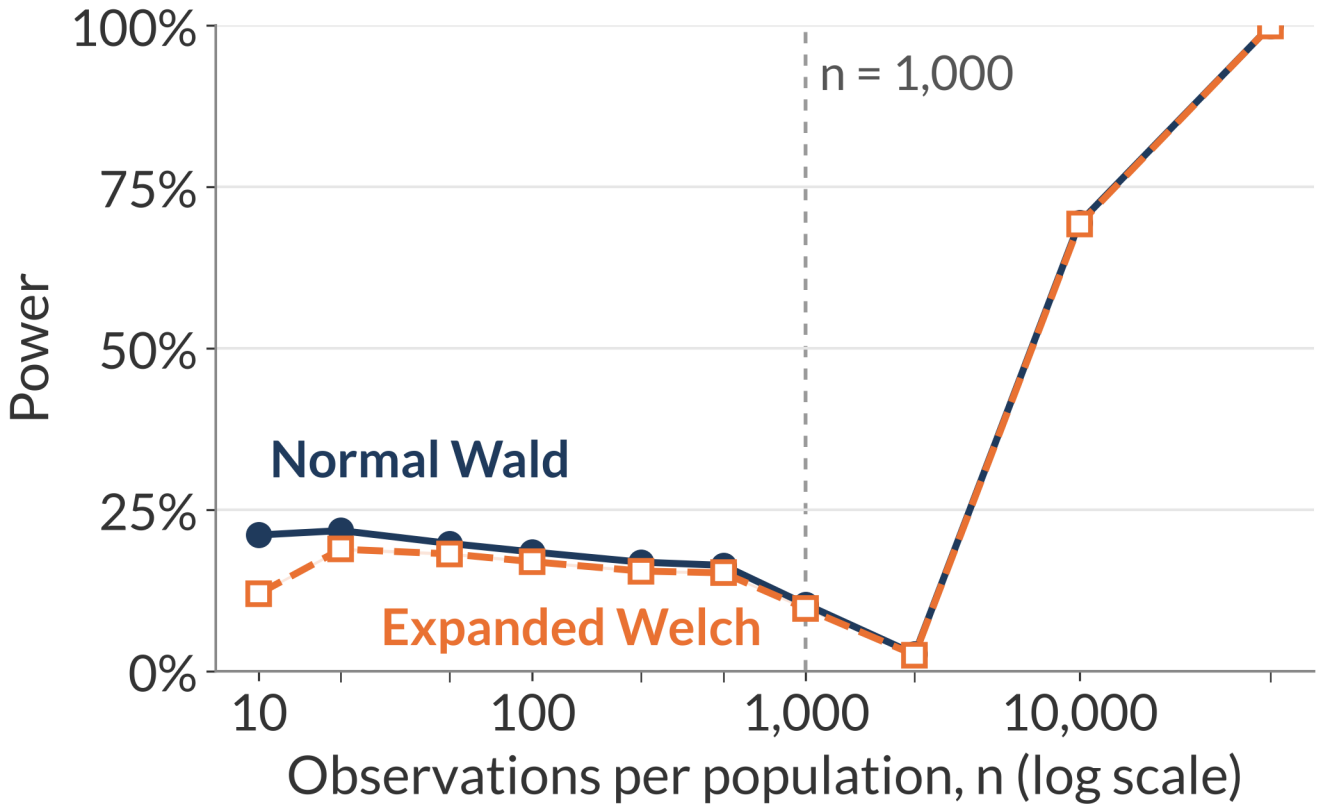
Bands: pointwise 95% Monte Carlo intervals. Thesis Section 6.3.1, Figure 6.7; full grids in B9–B11

Results: bias can reverse the estimated direction

In sparse tables, unequal residual biases can outweigh the true MI difference and reverse the estimated direction.

Different-skew 16×16 · MI 0.02 versus 0.03 nats

MI-gap magnitude 0.01 nats: power against sample size



At n = 1,000

True difference, P minus Q	**−0.0100 nats**
Mean estimated difference	**+0.0207 nats**

Expanded Welch rejects in **9.66%** of repetitions.

Among those rejections, over 99.9% have the opposite estimated direction.

Mechanism: As positive bias diminishes, the sampling centre passes near zero, so rejection initially falls before recovering.

At n = 1,000: 9,659 of 9,662 Expanded Welch rejections are positive; both methods 100% valid. Thesis Figure 6.9; full grid in B11

Illustration: online shopping data

Returning visitors have higher estimated month–purchase MI, and both approximate tests reject equal MI after adjustment.

Month × purchase: corrected MI estimates (nats)

New visitors

0.0043

Returning visitors

0.0179

Holm-adjusted p-values across six comparisons

Normal Wald 0.000189 **Expanded Welch** 0.00149

Both tests reject equal MI after Holm adjustment

Weekend × purchase

Estimates are close; neither test rejects equality.

Interpretation: Population MI is unknown. The simulations provide the evidence about reliability.

Sakar and Kastro (2018): 1,694 new and 10,551 returning sessions. Holm (1979). Thesis Section 6.4; details in B12

Conclusions and next steps

MI-specific degrees of freedom improve some false-positive rates, while remaining failures identify estimation accuracy as a priority.

Contribution and findings

1. Derived variance sensitivity and MI-specific degrees of freedom.
2. Brought some excessive false-positive rates closer to 5%; deepened conservatism elsewhere.
3. Reduced nominal power; residual bias can reverse the estimated direction.

Next steps

1. Improve bias and uncertainty estimates, including quadratic variation near independence.
2. Test other association constructions and unequal sample sizes.

Thank you.

Appendix

Variance sensitivity

Perturbation towards one cell: $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}$

Changing probability weights

$$g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P)$$

Changing pointwise-MI values and margins

$$+ 2[\ell_P(x,y) - \mathbb{E}_P\{\ell_P(X,Y) \mid X = x\}$$

$$- \mathbb{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)]$$

Pointwise MI: $\ell_P(i,j) = \log\{p_{ij}/(p_{i+}p_{+j})\}$

Sensitivity variance: $\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}$

Exact for positive population tables; its use for sampling errors is first order. Thesis Chapter 4

Moment matching and combined degrees of freedom

Component degrees of freedom

$$\nu_{P,\,\mathrm{mom}} = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}, \qquad \widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\hat{\tau}^2(P)}$$

Combined degrees of freedom

$$\widehat{\nu} = \frac{\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\}^2}{\{\widehat{V}(P)/n_P\}^2/\widehat{\nu}_P + \{\widehat{V}(Q)/n_Q\}^2/\widehat{\nu}_Q}$$

Moment matching does not establish the normality or independence needed for an exact Student t distribution.

Satterthwaite (1946). Thesis Sections 4.1 and 4.3

Population construction, grid and validity

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}\{i=j\}, \qquad 0 \le \lambda < 1$$

Parameter grid

- k = 2, 4, 8, 16; uniform, same-skew or different-skew margins
- Same skew: both use $m_k(0.8)$. Different skew: P uses $m_k(0.7)$, Q uses $m_k(0.8)$, where $m_k(\rho) = (\rho, (1-\rho)/(k-1), \ldots)$
- Null: common MI 0.005, 0.02, 0.30. Power: P MI 0.02, Q MI 0.025, 0.03, 0.04

Validity and rates

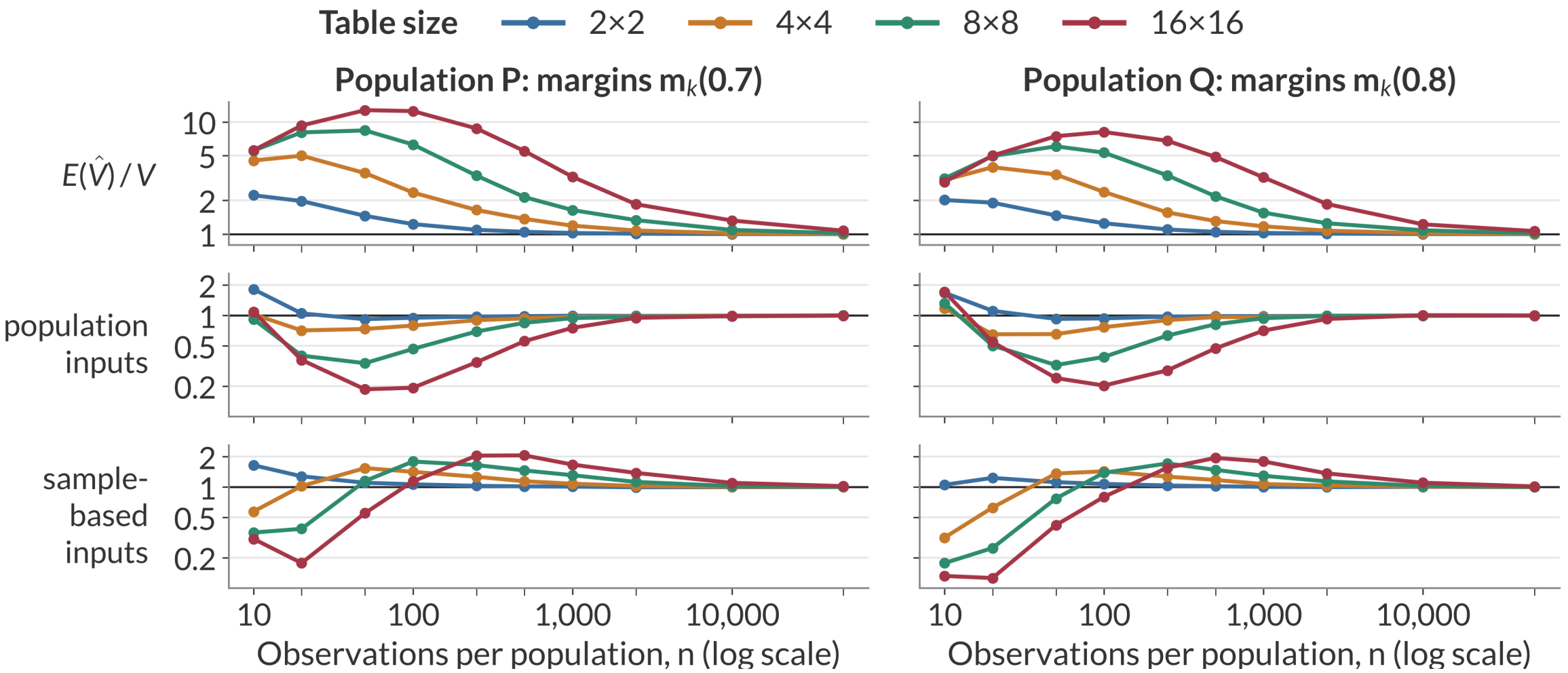
- Each study has 360 settings, 100,000 independent pairs per setting
- Wald needs a finite difference and positive finite standard error. Expanded Welch also needs valid positive finite df calculations.
- Rate = valid rejections / all repetitions. Report validity separately.

n = 10, 20, 50, 100, 250, 500, 1,000, 2,500, 10,000, 50,000 per group

Thesis Chapter 5 and Appendix B.1–B.5

Approximation ratios across table sizes

Different-skew margins, common MI 0.02 nats; a ratio of one indicates agreement



Rows: mean of $\hat{V}/V$; population-input and sample-based variance predictions / observed variance. Thesis Figure 6.1

Near-independence noise calculation

Setting: 4×4, n = 1,000, MI 0.0001; $d = 9$ and $\kappa = 2nI(P) \approx 0.2$

$$G = 2n\hat{I}(P) \;\dot{\sim}\; \chi^2_d(\kappa), \qquad \hat{\nu}_P \approx G/2$$

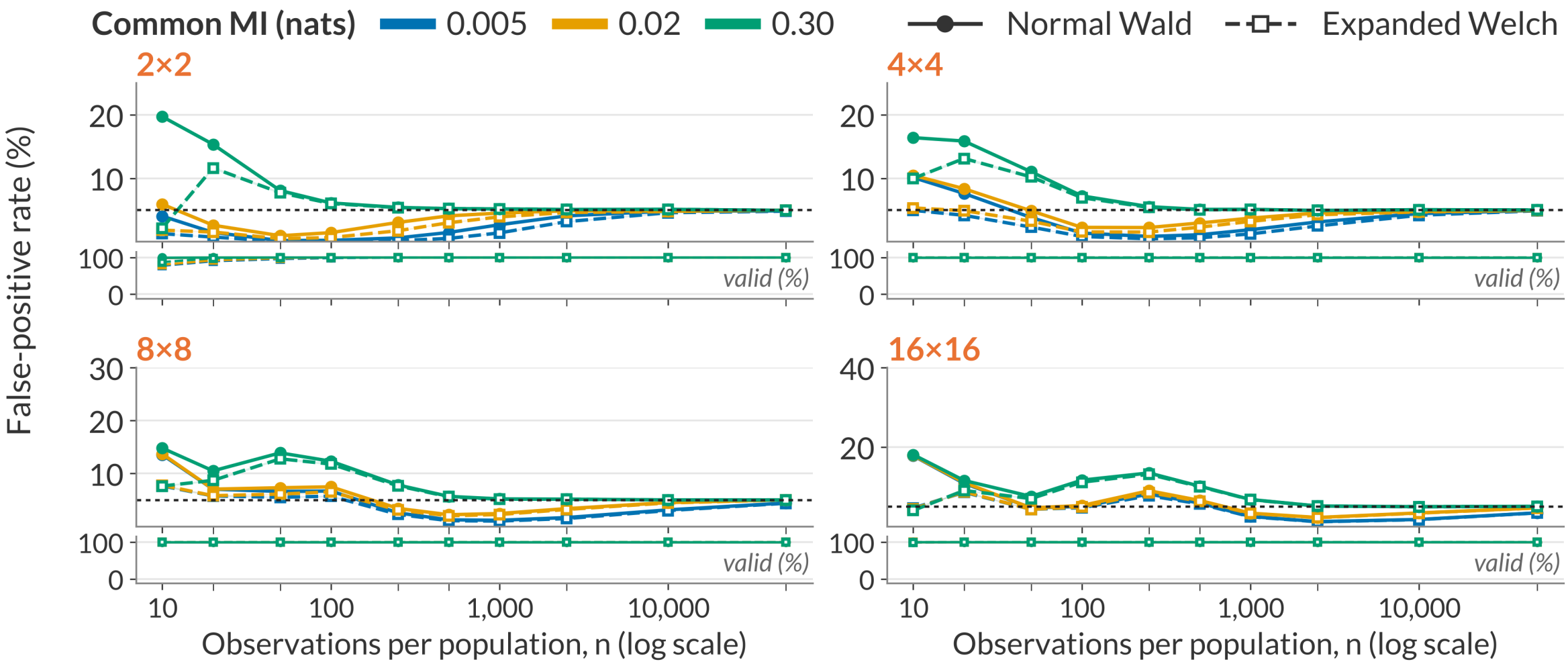
In $G, \hat{I}(P)$ is the uncorrected plug-in MI, not a bias-corrected estimate.

Quantity	Local prediction	Observed (P)
Mean estimated component df	4.60	4.86
Sampling-moment target	9.00	9.12
Mean sample-based variance prediction / observed sampling variance	1.96	1.97

Local approximations under fixed dimensions and positive limiting cell probabilities. Thesis Appendix A.2, Section 6.1.3

Full null results: uniform margins

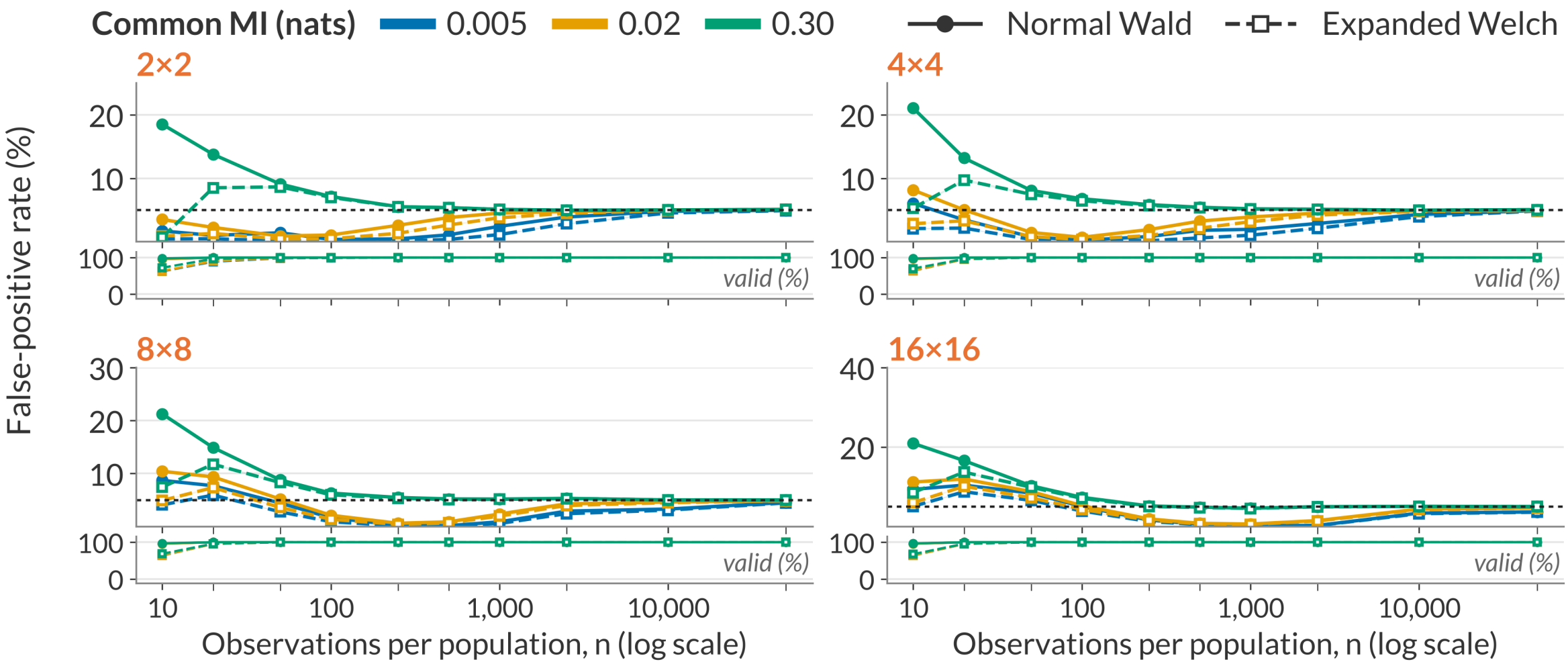
Equal MI in P and Q; uniform margins in both populations



Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.4

Full null results: same-skew margins

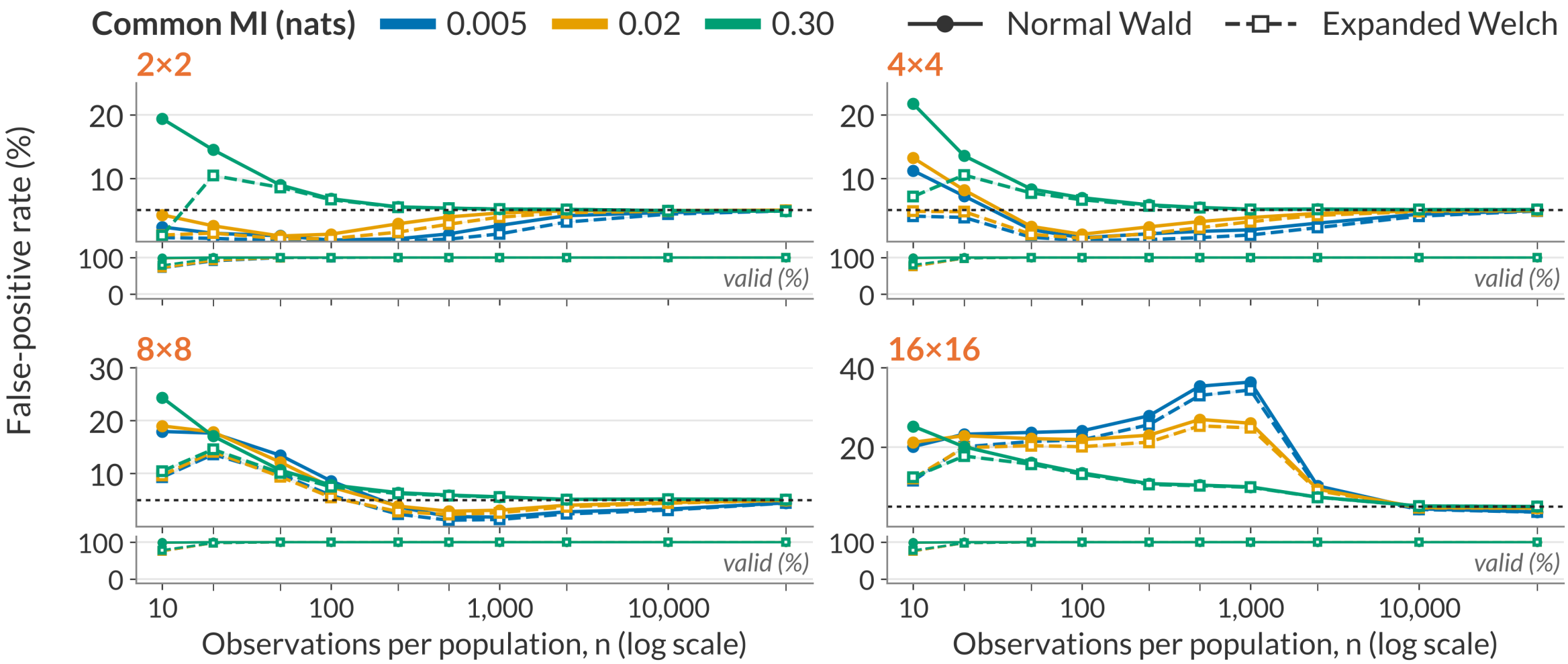
Equal MI; same-skew margins $m_k(0.8)$ in P and Q



Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.5

Full null results: different-skew margins

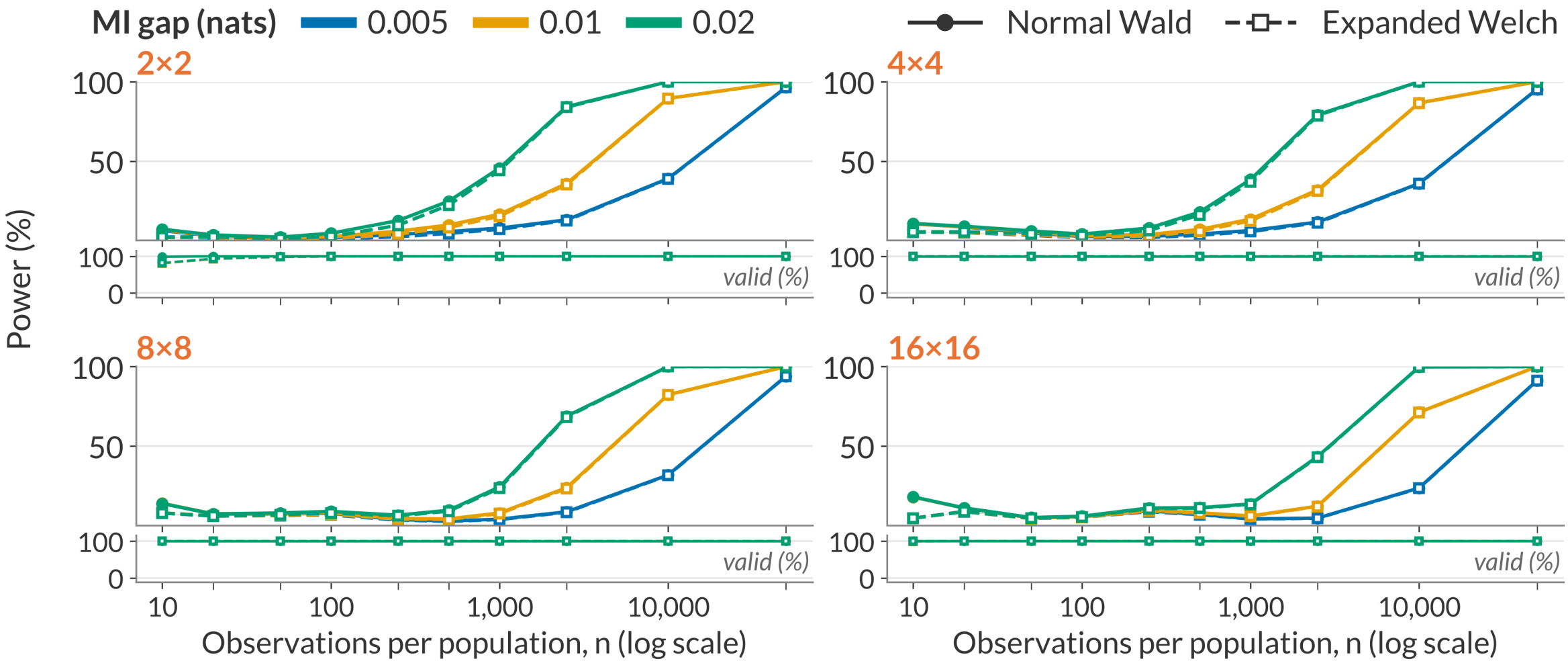
Equal MI; $m_k(0.7)$ in P and $m_k(0.8)$ in Q



Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.6

Full power results: uniform margins

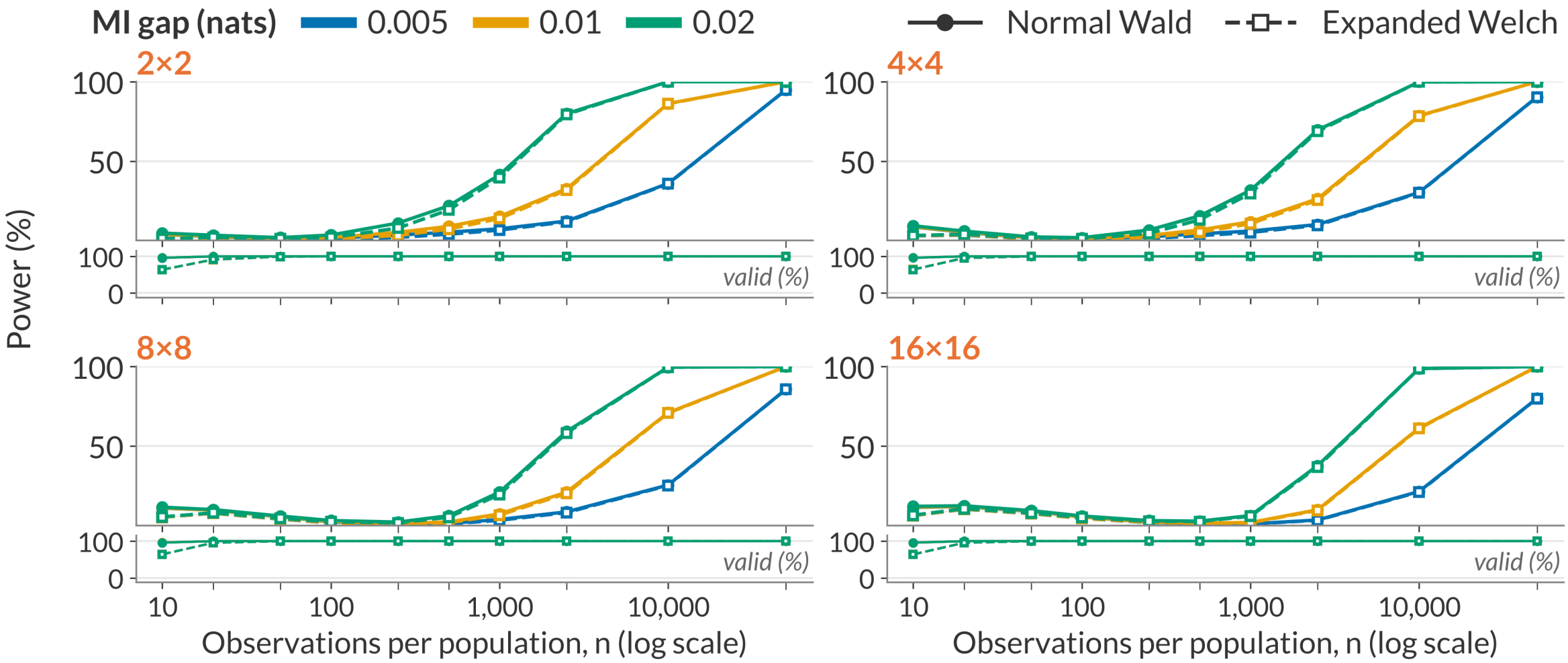
Uniform margins; MI in P = 0.02 nats



Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.7

Full power results: same-skew margins

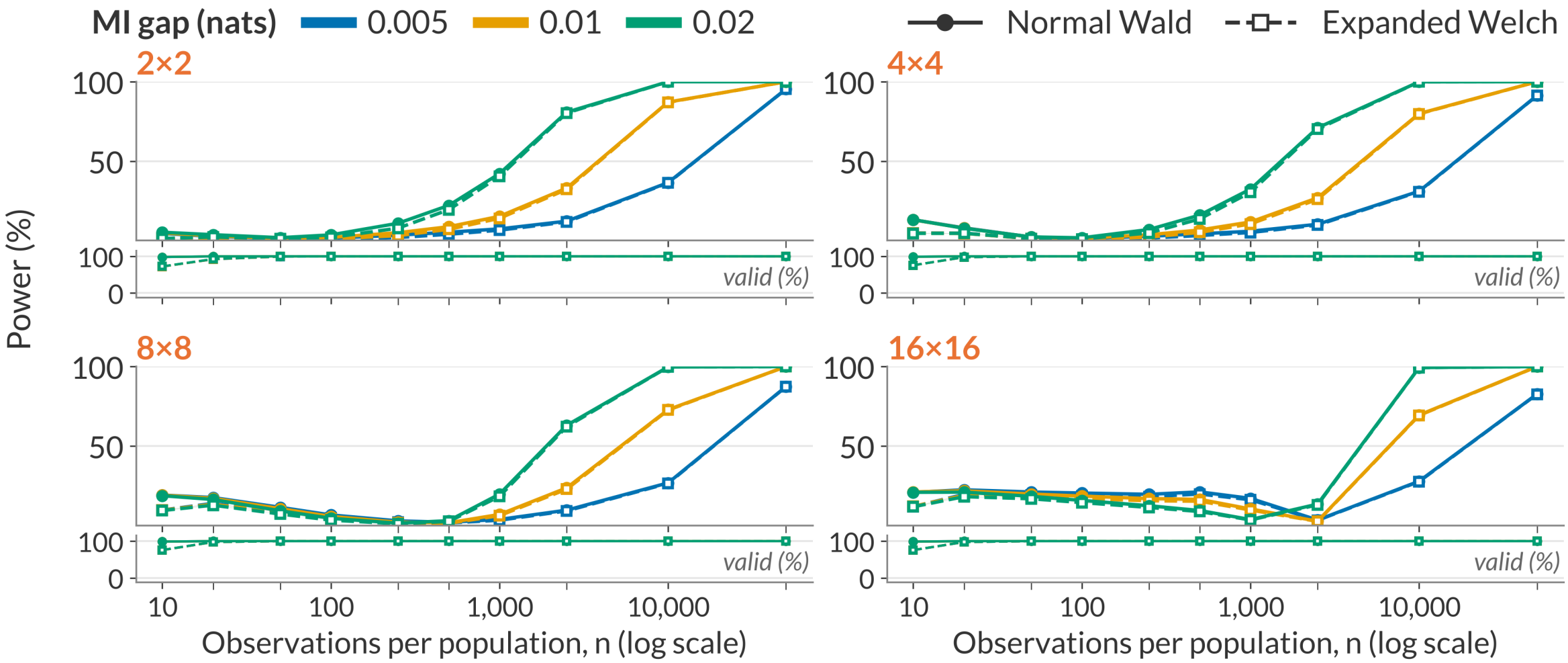
Same-skew margins $m_k(0.8)$ in P and Q; MI in P = 0.02 nats



Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.8

Full power results: different-skew margins

$m_k(0.7)$ in P, $m_k(0.8)$ in Q; MI in P = 0.02 nats



Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.9

Application details and comparison family

Unadjusted p-values for six exploratory comparisons; Holm adjusts each method separately

Dataset	Predictor	Normal Wald	Expanded Welch
Adult	Occupation	1.184×10^{-44}	1.635×10^{-42}
Adult	Education	1.890×10^{-34}	4.930×10^{-33}
Online Shoppers	Month	4.714×10^{-5}	3.726×10^{-4}
Student Performance	Study time	0.1982	0.2339
Student Performance	Maternal education	0.8361	0.8468
Online Shoppers	Weekend	0.9554	0.9601

Holm-adjusted month p-values: Normal Wald **0.000189**; Expanded Welch **0.00149**

Becker and Kohavi (1996); Sakar and Kastro (2018); Cortez (2008); Holm (1979). Thesis Appendix B.7, Table B.1

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